

Jordan–Wigner Transformation

From Second Quantization to Qubit Hamiltonians

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- Quantum computers operate with qubits (Pauli matrices)
- Many-body physics is naturally expressed in fermionic operators
- Need mapping:

$$c_p^\dagger, c_p \longrightarrow \text{Pauli operators}$$

Goal

Map a general second-quantized Hamiltonian to a sum of Pauli strings.

Fermionic Hamiltonian

General two-body Hamiltonian:

$$H = \sum_{pq} h_{pq} c_p^\dagger c_q + \frac{1}{2} \sum_{pqrs} V_{pqrs} c_p^\dagger c_q^\dagger c_s c_r$$

- One-body term: hopping / kinetic energy
- Two-body term: interaction

$$\{c_p, c_q^\dagger\} = \delta_{pq}, \quad \{c_p, c_q\} = 0$$

Challenge

Qubit operators commute on different sites:

$$[\sigma_i, \sigma_j] = 0 \quad (i \neq j)$$

Fermions anticommute!

Jordan–Wigner transformation

Define mapping:

$$c_j = \left(\prod_{k < j} z_k \right) \frac{X_j + iY_j}{2}$$

$$c_j^\dagger = \left(\prod_{k < j} z_k \right) \frac{X_j - iY_j}{2}$$

Key feature

Non-local string ensures correct fermionic anticommutation

[*citation* : 0Wikipedia](<https://en.wikipedia.org/wiki/Jordan>

$$n_j = c_j^\dagger c_j = \frac{1}{2}(1 - Z_j)$$

- Local operator
- No Jordan–Wigner string

One-body term

Consider:

$$c_p^\dagger c_q$$

Mapping gives:

$$c_p^\dagger c_q = \frac{1}{4} \left(\prod_{k=p}^{q-1} Z_k \right) (X_p X_q + Y_p Y_q + i X_p Y_q - i Y_p X_q)$$

Result

Becomes sum of Pauli strings with non-local Z string.

Two-body term

$$c_p^\dagger c_q^\dagger c_s c_r$$

- Each operator contributes:
 - Pauli matrices (X, Y)
 - Jordan–Wigner strings

Result

Transforms into a sum of Pauli strings of length $\mathcal{O}(N)$.

Total number of terms:

$$\mathcal{O}(N^4)$$

[citation : 1arXiv](https://arxiv.org/pdf/2505.08206?utm_source=chatgpt.com)

Structure after mapping

General form:

$$H = \sum_i c_i P_i$$

where:

$$P_i = \sigma_1^{\alpha_1} \otimes \sigma_2^{\alpha_2} \dots$$

- Pauli strings
- Non-local due to Z chains

Example: Nuclear pairing Hamiltonian

$$H = \sum_p \epsilon_p n_p - g \sum_{pq} c_p^\dagger c_{\bar{p}}^\dagger c_{\bar{q}} c_q$$

- Pair creation:

$$P_p^\dagger = c_p^\dagger c_{\bar{p}}^\dagger$$

Mapping pairing operators

Using Jordan–Wigner:

$$c_p^\dagger c_{\bar{p}}^\dagger \rightarrow \text{Pauli strings with } X, Y, Z$$

Important simplification

Time-reversed pairs often cancel parts of the JW string.

Result:

$$H = \sum_p \epsilon_p \frac{1 - Z_p}{2} - g \sum_{pq} (X_p X_q + Y_p Y_q + \dots)$$

General two-body interaction

$$c_p^\dagger c_q^\dagger c_s c_r$$

- Produces:
 - XXXX
 - YYXX
 - mixed terms

Key observation

Interaction terms generate entangling Pauli strings.

- Number of Pauli terms:

$$\mathcal{O}(N^4)$$

- String length:

$$\mathcal{O}(N)$$

Implication

Jordan–Wigner is simple but produces non-local operators.

Physical interpretation

- Z -strings encode fermionic parity
- X, Y encode creation/annihilation

$$\prod_{k < j} Z_k = (-1)^{\text{fermions to the left}}$$

Meaning

This enforces the Pauli exclusion principle.

Advantages and limitations

Advantages

- Simple
- Exact

Limitations

- Non-local strings
- Long circuits

Summary

- Jordan–Wigner maps fermions to qubits
- General Hamiltonians become sums of Pauli strings
- Pairing Hamiltonians simplify due to symmetry
- Two-body interactions produce entangling terms

Final insight

Fermionic statistics $\rightarrow$ non-local qubit operators